

# Dose-dependent effects of inhaled anesthetics on organ systems

|                                | Isoflurane                                                                                                                                                                                                                                       | Desflurane | Sevoflurane | Nitrous Oxide                                                                     |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|-----------------------------------------------------------------------------------|
| Cardiovascular<br>(next slide) | ↓ blood pressure, little to no effect on cardiac output<br>Depress baroreflex (sevoflurane > isoflurane and desflurane)<br><i>Desflurane: sympathetic stimulation → tachycardia, HTN</i><br>Isoflurane / Sevoflurane → ↑HR with increasing doses |            |             | Negligible effects<br>(mild ↑ SNS stimulation that offsets myocardial depression) |
| Respiratory                    | Respiratory depression ( <i>profound with isoflurane</i> )<br>↓ tidal volume and minute ventilation<br>↓ ventilatory response to carbon dioxide<br>Bronchodilation ( <i>most pronounced with sevoflurane</i> )                                   |            |             | Minimal change in minute ventilation and resting arterial pCO <sub>2</sub>        |
| Cerebral                       | ↑ cerebral blood flow and ↑ intracranial pressure*<br>↓ cerebral metabolic rate<br><i>Sevoflurane <del>decreases</del> increases CBF and ICP to lesser extent.</i>                                                                               |            |             | ↑CBF, ↑ICP<br>↑ cerebral metabolic rate                                           |
| Neuromuscular                  | Relax skeletal muscle<br>Enhance the effects of neuromuscular blockers                                                                                                                                                                           |            |             | Does not relax skeletal muscle or smooth muscle                                   |
| Smooth muscle                  | Relax smooth muscle → vasodilation, bronchodilation, uterine smooth muscle relaxation                                                                                                                                                            |            |             |                                                                                   |
| Renal                          | ↓ renal blood flow, ↓ glomerular filtration rate, ↓ urine output                                                                                                                                                                                 |            |             | The same                                                                          |
| Hepatic                        | ↓ hepatic blood flow                                                                                                                                                                                                                             |            |             | The same                                                                          |
| Hepatic metabolism             | 0.2%                                                                                                                                                                                                                                             | <0.1%      | 5%          | 0.004%                                                                            |

\*↑ Intracranial pressure: Caution in patients with brain tumor, intracranial hemorrhage, or head injury. Anesthetic-related increase in the ICP can cause potentially devastating neurological complications.



## Hemodynamic effects of inhaled anesthetics (dose-dependent)

| Effects                                                                                                                                                                                                    | Comments                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Depression of cardiac contractility                                                                                                                                                                        | Decrease mean arterial pressure in direct proportion to the alveolar concentration of the gas                                                                                                                                                                                                                                                  |
| Vasodilation                                                                                                                                                                                               | <div><div><p><b>Halothane</b> / enflurane more myocardial depression than the others, minimal vasodilation</p><p>↓</p><p>↓ CO with little change in SVR</p></div><div><p><b>Isoflurane, desflurane, sevoflurane</b></p><p>↓</p><p>Vasodilation (↓BP) / minimal change in CO</p><p>↓</p><p>Preserve CO + ↓ afterload, ↓ preload</p></div></div> |
| <b>Takeaway:</b> Isoflurane, desflurane, and sevoflurane have minimal myocardial depressant effects and favorable hemodynamic profiles and may be suitable for patients with impaired myocardial function. |                                                                                                                                                                                                                                                                                                                                                |
| Desflurane                                                                                                                                                                                                 | <div><p>Transient sympathetic stimulation → tachycardia and hypertension, especially at high doses. Tachycardia may persist.</p><p><b><i>Use with caution and close monitoring in patients with coronary artery disease.</i></b></p></div>                                                                                                     |

CO: cardiac output; SVR: systemic vascular resistance

➤ At high doses that induce Stage IV anesthesia, **all agents** can cause cardiovascular collapse.



# Emergence and Postoperative Phenomena

| Effects                             | Comments                                                                                                                                                                                                                                                                                                |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nausea and vomiting</b>          | Drug-induced stimulation of chemoreceptor trigger zone and vomiting center<br>Management:<br>1) antiemetics: metoclopramide, ondansetron, droperidol<br>2) avoidance of nitrous oxide (N <sub>2</sub> O)<br>3) use of propofol as an induction agent<br>4) use of ketorolac as a substitute for opioids |
| <b>Hypertension and tachycardia</b> | ↑ Sympathetic tone<br>Sympathetic tone is enhanced by pain.                                                                                                                                                                                                                                             |
| <b>Post-anesthesia shivering</b>    | Result of decrease core temperature (hypothermia) and neurological after-effects of the general anesthetic agent.                                                                                                                                                                                       |
| <b>Emergence excitement</b>         | Tachycardia, restlessness, crying, moaning, and thrashing, and a variety of neurologic signs, including delirium, spasticity, hyperreflexia, and Babinski signs                                                                                                                                         |



## Descriptions of Toxicities

| Toxicity               | Isoflurane                                                                                                                                                                                                                                                                                                                                                | Desflurane | Sevoflurane                                        | Halothane | Enflurane          | Nitrous oxide |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------|-----------|--------------------|---------------|
| Acute                  | Narrow therapeutic range<br>Acute organ damage is low when used by trained practitioners                                                                                                                                                                                                                                                                  |            |                                                    |           |                    |               |
| Malignant hyperthermia | Individuals with genetic disorders of skeletal muscle can develop muscle rigidity, hyperthermia, rapid onset of tachycardia and hypercapnia, hyperkalemia, and metabolic acidosis when exposed to <b>succinylcholine</b> or <b>halogenated anesthetic</b> agents                                                                                          |            |                                                    |           |                    | —             |
| Nephrotoxicity         | —                                                                                                                                                                                                                                                                                                                                                         | —          | compound A vinyl ether<br>No renal injury reported | —         | Prolonged exposure | —             |
| Hepatotoxicity         | <p>➤ Hepatic dysfunction following surgery and general anesthesia is most likely caused by hypovolemic shock, infection conferred by blood transfusion, or other surgical stresses rather than by volatile anesthetic toxicity.</p> <p>Halothane: Fulminant liver failure (1 in 20,000 to 35,000) by direct toxicity and immune response.<sup>7</sup></p> |            |                                                    |           |                    |               |

|                            | Isoflurane                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Desflurane | Sevoflurane | Halothane | Enflurane | Nitrous oxide |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|-----------|-----------|---------------|
| Neurotoxicity              | <p>Neural toxicity in infants and young children is an emerging concern. Some, not all, studies suggest increased risk of neurodevelopmental deficits in motor function and behavior.</p> <p><i>Inadequate evidence to directly attribute the effects to the anesthesia.</i></p>                                                                                                                                                                               |            |             |           |           |               |
| Carbon monoxide production | <p>All halogenated anesthetics can produce some carbon monoxide (CO) from their interaction with strong bases in <b>desiccated (dried out) carbon dioxide absorbers</b>. CO binds with hemoglobin with high affinity, reducing oxygen delivery to tissues.</p> <p>Prevention: Fresh carbon dioxide absorbent / avoid dessication</p> <p><i>Potential for CO formation from most to least:</i><br/> <i>desflurane &gt; isoflurane &gt; &gt; sevoflurane</i></p> |            |             |           |           | —             |
| See slide 47               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |             |           |           |               |
| Carcinogenicity            | <p>Under normal conditions, inhaled anesthetics are neither mutagens nor carcinogens. No study has demonstrated a causal relationship between anesthetics and cancer.</p>                                                                                                                                                                                                                                                                                      |            |             |           |           |               |
| Pregnancy                  | <p>In the US, at least 50,000 pregnant women each year undergo anesthesia and surgery for indications unrelated to pregnancy. The risk of abortion is clearly higher following this experience. It is not obvious whether the underlying disease, surgery, anesthesia, or a combination of these factors is the cause of the increased risk.</p>                                                                                                               |            |             |           |           |               |
| Environment                | <p>All inhaled anesthetics are greenhouse gases – hundreds of thousands of times more potent than CO<sub>2</sub> – They are vented out the buildings into the environment.</p>                                                                                                                                                                                                                                                                                 |            |             |           |           |               |

## Nitrous oxide's drug-specific effects

|                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Body space volume expansion                                         | <p><math>N_2O</math> is ~35x more soluble in blood than <math>N_2</math> → <math>N_2O</math> diffuses into cavities faster than <math>N_2</math> diffuses out → increases the volume inside the cavity → expansion of the cavity</p> <p><b>When walls surrounding the cavity are rigid, the pressure rises instead of the volume.</b></p>                                                                                                                                             | <p>Cautions:</p> <p>Pneumothorax   Intracranial air</p> <p>Middle ear surgery   Bowel obstruction</p> <ul style="list-style-type: none"> <li><math>N_2O</math> will also diffuse into tracheal tube cuffs, increasing the pressure against the tracheal mucosa.</li> </ul> |
| Vitamin B12 inhibition                                              | <p>Nitrous oxide interferes with vitamin B12 cofactor functions → inactivation of methionine synthase →</p> <ul style="list-style-type: none"> <li>Signs of vitamin B12 deficiency   megaloblastic anemia   peripheral neuropathy</li> </ul> <p><b><math>N_2O</math> should not be used for chronic pain relief (analgesia) or as a sedative in critical care settings.</b></p> <p><b>Avoid in patients with disorders of B12, folate, or methionine synthesis or metabolism.</b></p> | <p>Biochemistry: Vitamin B12 is a cofactor of methionine synthase, which generates methionine from homocysteine.</p> <p>Methionine is necessary for the formation of tetrahydrofolate, which is required for the synthesis of DNA precursors.</p>                          |
| Diffusion hypoxia                                                   | <p>On discontinuation of <math>N_2O</math> administration, <math>N_2O</math> diffuses from the blood to the alveoli, which dilutes the oxygen in alveoli.</p>                                                                                                                                                                                                                                                                                                                         | <p>Prevention: 100% <math>O_2</math> administration instead of air when <math>N_2O</math> is discontinued</p>                                                                                                                                                              |
| Second gas effect<br>The concentration effect of one gas on another | <p>Rapid uptake of <math>N_2O</math> from the alveoli → contraction of alveolar volume associated with the <math>N_2O</math> uptake → concentrates the co-administered inhaled anesthetic remaining in the alveoli → speeds induction</p>                                                                                                                                                                                                                                             | <p>Probably insignificant in the clinical practice of anesthesiology</p>                                                                                                                                                                                                   |



## Check your knowledge

What is the acronym MAC stand for?

How does MAC define potency?

How is MAC used in dosing inhaled anesthetics?

Which inhaled anesthetics are potent volatile agents in common use?

Why is nitrous oxide referred to as a weak anesthetic?

## Check your knowledge

Why may isoflurane, desflurane, and sevoflurane be suitable for patients with impaired myocardial function?

What is the concern with the use of desflurane in patients with coronary artery disease?

Which inhaled anesthetics are associated with malignant hyperthermia in patients with a genetic disorder in skeletal muscle calcium regulation?



## Check your knowledge

Match the toxicities with the drugs that cause them

|                                                        |
|--------------------------------------------------------|
| Body space volume expansion                            |
| Carbon monoxide production in desiccated CO2 absorber  |
| Cortisol synthesis inhibition                          |
| Diffusion hypoxia                                      |
| Environmental toxicity                                 |
| Malignant hyperthermia                                 |
| Myoclonus                                              |
| Nausea and vomiting                                    |
| Nephrotoxicity (potentially) by compound A             |
| Sympathetic stimulation → tachycardia and hypertension |
| Vitamin B12 inhibition                                 |

|               |
|---------------|
| Isoflurane    |
| Desflurane    |
| Sevoflurane   |
| Nitrous oxide |
|               |
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|               |
|               |

# Redistribution and intravenous anesthetic's duration of effect

## Single bolus dose:

### 1. Rapid onset:

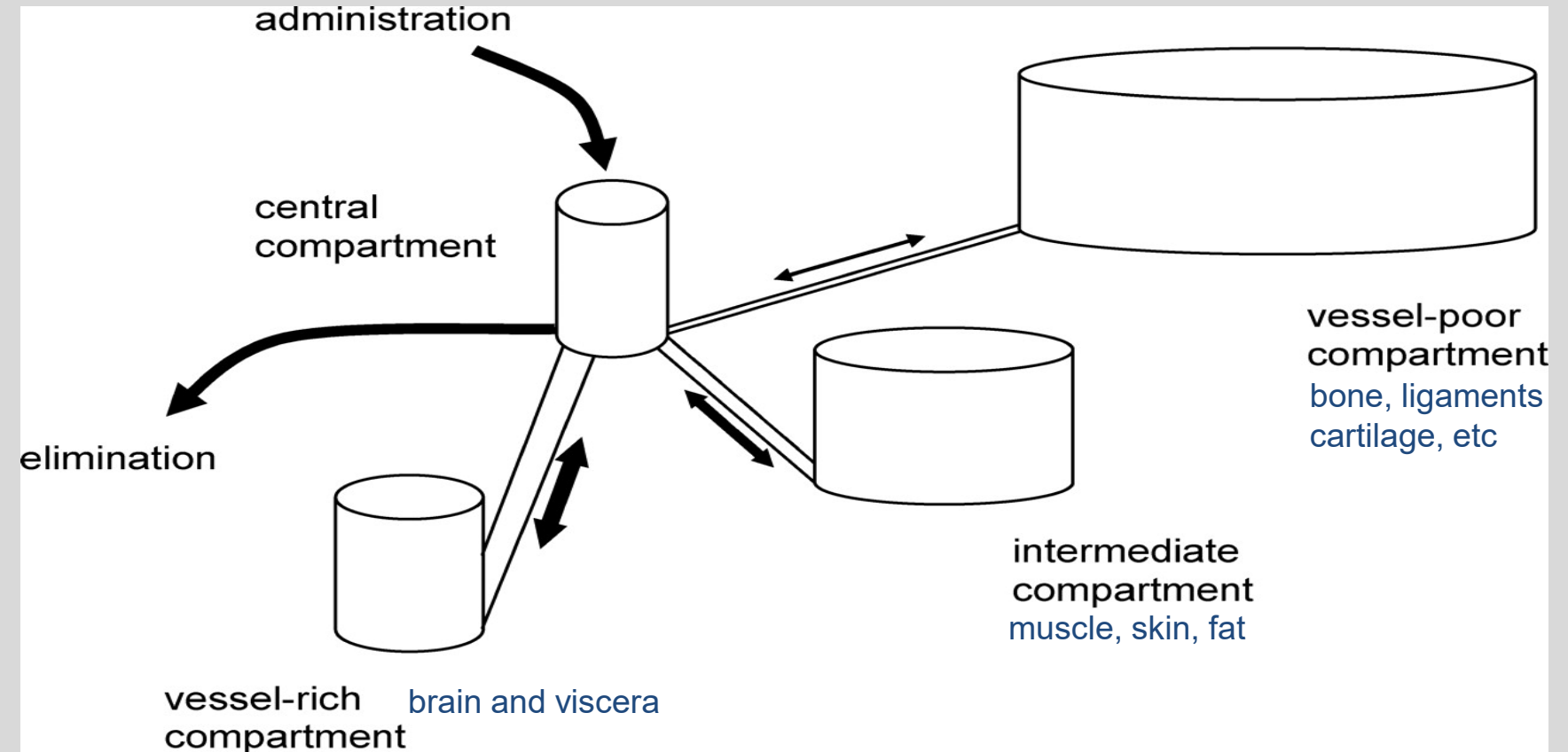
Highly lipophilic IV anesthetics partition into the highly perfused and lipophilic tissues of the brain and spinal cord → anesthesia in a single circulation time.

### 2. Termination of effect:

As the drug concentration in blood decreases, the anesthetic begins to redistribute out of CNS to less perfused tissues, diminishing the effect → patient emerges from anesthesia

### 3. Elimination:

Eventually, the drug is metabolized by the liver and excreted by the kidneys.



Three compartment model for a lipid soluble drug.

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## Context-sensitive half-time

**Half-time:** The time required for the concentration of drug to decrease by 50% after stopping a continuous infusion.

**Context:** The duration of the infusion

1. IV anesthetic initially distributes to the well perfused tissues.
  2. The drug redistributes to the less perfused tissues over time.
  3. The longer the infusion, the more the drug accumulates in the muscle and fat compartments.
  4. After the infusion is stopped and the drug is metabolized and excreted, blood levels begin to decrease.
  5. The drug redistributes from the peripheral compartments back into the bloodstream.
- For drugs with a long infusion duration, the context-sensitive half-time (time it takes for the concentration of drug in the blood to decrease by one-half) can be significantly longer due to extensive accumulation of drug in peripheral compartments.

Clinical importance:

Context-sensitive half-time helps predict how long it will take the patient to recover after the infusion is stopped.



## I.V. Anesthetics Mechanisms of action and Uses

| Drug      | Advantages                                                                                                                                                                                                                                                             | Disadvantages                                                                                                                                                                                                                                                                                                                                                       |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Propofol  | Bronchodilation<br>Decreases CMRO <sub>2</sub> , CBF, ICP<br>Antiemetic / antipruritic<br>Anticonvulsant                                                                                                                                                               | Hypotension, dose related<br>Respiratory depression, dose related<br>*Propofol infusion syndrome<br>Painful injection<br>Egg phosphate in soybean oil vehicle (looks like bottled milk):<br>Potential allergens and microbial risk                                                                                                                                  |
| Etomidate | Hemodynamic stability; no changes in BP, HR, or CO<br>Decreases CMRO <sub>2</sub> , CBF, ICP<br>Anticonvulsant                                                                                                                                                         | High incidence of postop nausea/vomiting<br>Painful injection<br>No analgesic effects<br>Involuntary myoclonic movements<br>Transient acute adrenal suppression<br>Inhibits 11-β-hydroxylase → ↓ cortisol synthesis                                                                                                                                                 |
| Ketamine  | Profound analgesic properties<br>Bronchodilation<br>Sympathetic stimulation → maintains CO and MAP without inhibiting respiratory drive (may be beneficial in cardiac instability)<br>Maintains airway reflexes and respiratory drive<br>IM route if IV access is lost | Sympathetic stimulation increases HR, BP, CO and myocardial O <sub>2</sub> demand (may be harmful in CAD or HTN),<br>Increase pulmonary arterial pressure SBP<br>Potentiates cardiovascular toxicity of cocaine, tricyclic antidepressants<br>Increases CBF, ICP<br>Hallucinations, delirium upon withdrawal, dissociative experiences, nightmares, nausea/vomiting |

CMRO<sub>2</sub>: cerebral metabolic rate of oxygen; CBF: cerebral blood flow; ICP: intracranial pressure; BP: blood pressure; HR: heart rate; CO: cardiac output; MAP: mean arterial pressure; SBP: systemic blood pressure; CAD: coronary artery disease; HTN: hypertension

\*Propofol infusion syndrome, a rare complication: metabolic acidosis, hypotension, hepatomegaly, hyperlipidemia, rhabdomyolysis, CV collapse

## Summary of Inhaled Anesthetics

| Anesthetic    | B:G  | B:B | <sup>1</sup> MAC | Comments                                                                                                                                                                                                                                                    |
|---------------|------|-----|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nitrous oxide | 0.47 | 1.1 | >100             | Weak anesthetic; rapid onset and recovery<br>Not metabolized<br>Can cause expansion of air-fill cavities / Diffusional hypoxia<br>Prolonged exposure can result in megaloblastic anemia, peripheral neuropathies                                            |
| Desflurane    | 0.42 | 1.3 | 6-7              | Low volatility; rapid onset/recovery; poor induction agent (respiratory irritation)<br>SNS stimulation → transient, often marked, tachycardia, ↑BP<br>Carbon monoxide can be produced from desiccated CO <sub>2</sub> absorbents in breathing circuit       |
| Sevoflurane   | 0.69 | 1.7 | 2.0              | Rapid onset/ recovery<br>Smooth rapid inhalation induction; bronchodilating – preferred for patients with airway problems<br>Hepatic metabolism (~5% of absorbed drug)<br>Interaction with soda-lime → Compound A vinyl ether (renal toxicity not reported) |
| Isoflurane    | 1.40 | 2.6 | 1.4              | Medium rate of onset and recovery; pungent odor → poor induction agent; dilates coronary arteries                                                                                                                                                           |
| Enflurane     | 1.80 | 1.4 | 1.7              | Medium rate of onset / recovery; limited biodegradation; seizure potential                                                                                                                                                                                  |
| Halothane     | 2.30 | 2.9 | 0.7              | Relatively slow onset / recovery; significant hepatic metabolism; halothane hepatitis – reactive intermediary – direct toxicity, immune reaction                                                                                                            |

<sup>1</sup>MAC=median alveolar concentration in vol%: the anesthetic concentration that produces immobility in 50% of patients exposed to noxious stimulus; MAC is small for potent anesthetics and large for less potent anesthetics.

B:G=blood:gas partitions coefficients; B:B=brain:blood partition coefficients

Adapted from Basic & Clinical Pharmacology Table 25-2



## Common advantages and disadvantages of potent volatile anesthetics

### Advantages

- Ease of administration
- Reliable block of sensory input (ie, pain)
- Bronchodilation
- Decrease in skeletal and smooth muscle tone (dose-dependent)
- Decreased cerebral metabolic rate
- Increased cerebral blood flow by direct vasodilatory effect
- Monitoring end-tidal anesthesia concentration as a measure of anesthetic depth

### Disadvantage

- Dose-dependent...
  - Suppression of airway reflexes
  - Respiratory depression
  - Hypotension due to myocardial depression and vasodilation
- Increased post-op nausea/vomiting compared to IV anesthetics (antiemetic needed)
- Malignant hyperthermia potential in susceptible individuals



## Nitrous Oxide: Summary of effects

|                    |                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anesthetic effects | <p>Incomplete (weak) anesthesia</p> <ul style="list-style-type: none"> <li>➤ Adjunctive use with inhalation and intravenous anesthetics for sedation, analgesia, and anxiolytic effects</li> </ul>                                                                                                                                                                |
| Muscle effects     | <p>Does not relax skeletal muscle → does not enhance the effects of neuromuscular blocking drugs</p> <p>Does not relax uterine smooth muscle</p> <ul style="list-style-type: none"> <li>➤ Does not cause malignant hyperthermia</li> </ul>                                                                                                                        |
| Organ effects      | <p>Sympathetic stimulation → maintains hemodynamics</p> <p>Increases respiratory rate</p> <p>Reduces hepatic and renal blood flow   not hepato- or nephrotoxic</p>                                                                                                                                                                                                |
| Other effects      | <ul style="list-style-type: none"> <li>• Body space volume expansion and increased pressure in spaces with noncompliant walls.</li> <li>• Interferes with vitamin B12 function; avoid in patients with disorders of B12 deficiency and folate or methionine synthesis</li> <li>• Diffusion hypoxia</li> <li>• Second gas effect (concentration effect)</li> </ul> |





| IV Agents             | MOA                                                 | Benefits                                                                                                                                                       | Adverse effects                                                                                                                       | Notes                                                                                                          |
|-----------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Midazolam             | GABA-A positive allosteric modulator                | Sedation<br>Anxiolytic<br>Anterograde amnesia<br>Antiseizure                                                                                                   | Mild vasodilation, ↓BP and cardiac output<br>Respiratory depression with concurrent opioid                                            | Not frequently used for deep sedation – propofol and etomidate have superior effectiveness                     |
| Opioids               | Mu opioid receptor agonists                         | Analgesia<br>Supplemental sedation<br>Suppress airway reflexes during intubation<br>Reduce stress responses                                                    | Respiratory depression<br>Bradycardia<br>Exacerbate hypotension by sedative-hypnotic<br>Nausea, vomiting, ileus                       | Common use<br>Short-acting opioids during induction                                                            |
| Lidocaine IV infusion | Blocks voltage-gated sodium channels in nerve cells | Suppress the cough reflex laryngospasm<br>Reduces pain of injection or other induction agents, and supplement anesthetic effects<br>Decreases need for opioids | Lidocaine-induced ventricular rate can contribute to arrhythmia. Arrhythmias are uncommon when IV lidocaine is used for pain control. | Lidocaine infusion requires close monitoring for signs of cardiac and local anesthetic systemic toxicity(LAST) |

## Check your knowledge

What is the acronym MAC stand for?

- Minimum alveolar concentration

How does MAC define potency?

- 1 MAC = the concentration of the inhaled gas in the alveoli that prevents movement in 50% of patients in response to a standardized stimulus (eg, surgical incision) at standard temperature and ambient pressure.
- MAC reflects brain partial pressure and allows comparisons of potency between agents.

How is MAC used in dosing inhaled anesthetics?

- Dosing of inhaled anesthetics is guided by MAC and MAC values are additive.

Which inhaled anesthetics are potent volatile agents in common use?

- Isoflurane, desflurane, sevoflurane

Why is nitrous oxide referred to as a weak anesthetic?

- Nitrous oxide has low potency. Even 100% nitrous oxide does not achieve 1 MAC. Therefore, it is combined with a potent volatile anesthetic.

## Check your knowledge

Why may isoflurane, desflurane, and sevoflurane be suitable for patients with impaired myocardial function?

- Isoflurane, desflurane, and sevoflurane cause vasodilation, which lowers systemic vascular resistance and blood pressure with minimal change in cardiac output (preserves cardiac output) – of course, with close monitoring to ensure patient safety and optimal outcomes.

What is the concern with the use of desflurane in patients with coronary artery disease?

- Although desflurane causes vasodilation with minimal effect on the force of contraction (inotropy), its transient sympathomimetic properties cause tachycardia and hypertension, especially at high dose. These cardiac effect may induce cardiac ischemia and heart attack.

Which inhaled anesthetics are associated with malignant hyperthermia in patients with a genetic disorder in skeletal muscle calcium regulation?

- The halogenated anesthetics: isoflurane, desflurane, sevoflurane, halothane, enflurane
- Not nitrous oxide.

## Check your knowledge

Match the toxicities with the drugs that cause them

|                                                        |                                  |
|--------------------------------------------------------|----------------------------------|
| Body space volume expansion                            | Nitrous oxide                    |
| Carbon monoxide production in desiccated CO2 absorber  | Halogenated volatile anesthetics |
| Cortisol synthesis inhibition                          | Isoflurane                       |
| Diffusion hypoxia                                      | Nitrous oxide                    |
| Environmental toxicity                                 | All inhaled anesthetics          |
| Malignant hyperthermia                                 | Halogenated volatile anesthetics |
| Myoclonus                                              | Isoflurane                       |
| Nausea and vomiting                                    | All inhaled anesthetics          |
| Nephrotoxicity (potentially) by compound A             | Sevoflurane                      |
| Sympathetic stimulation → tachycardia and hypertension | Desflurane                       |
| Vitamin B12 inhibition                                 | Nitrous oxide                    |